Planning as a Cognitive Cost that Discourages Goal Switching

Luke Cho
Princeton Neuroscience Institute, Princeton University

J. Branson Byers
Department of Psychology, Princeton University

Yael Niv
Princeton Neuroscience Institute, Princeton University
Department of Psychology, Princeton University

Abstract

People exhibit a striking preference for maintaining their current goals. While this bias can be framed as irrational (e.g., the sunk cost fallacy), this persistence may be adaptive for conserving cognitive resources. We hypothesized that humans are sensitive to the cost of planning that comes with switching between goals. We conducted a behavioral experiment ($n = 60$) in which participants pursued goal completion under two conditions: one in which they had to plan towards goals before pursuing them and another in which they were provided with a pre-specified optimal plan. We found that people persisted with their goals more when they had to create their own plans. We quantified the cognitive costs of planning with behavioral modeling and found that planning costs predict decreased goal switching. These results indicate that individuals are sensitive to the cost of planning and suggest that goal persistence may be an adaptive response to these cognitive costs.